

Five Walls on the Path to Growing a Grounded Language: An Adversarial Self-Audit of Emergent Communication

Chunxuan Yang

Sogang University, Seoul, Republic of Korea
yang2023@sogang.ac.kr yangchunxuan1@gmail.com

Abstract

A standing goal of artificial life is a population of from-scratch agents that *grows* a shared, decodable, grounded language that is *load-bearing* for survival and cooperation. **We do not solve this**; we identify why several plausible routes fail, and contribute a methodology that prevents such failures from being misreported as success. Every candidate “win” must survive four checks together: an architecture-matched null, a fair baseline, ≥ 6 seeds with a bootstrap confidence interval on the win-margin, and an adversarial red-team. This rig caught *six* of our own apparent wins as artifacts. Across five distinct attacks we banked zero clean breaks: each fell to a cheap shortcut the world allowed, suggesting that *in our tested regimes world design was a recurring bottleneck*, and that structure does not grow in worlds too small to require it. We then constructively build a world (a Relay-Tide Commons) that closes each shortcut. Inside it, grown-channel content is *relatively* load-bearing for survival against degraded controls, and its *use* is selected up under evolution—but we detect *no* co-evolution of the channel itself from random initialization under survival selection (an apparent intelligibility rise did not separate from a communication-blind control; $n=6$, underpowered). We offer the self-audit protocol, the mapped wall system, and reproducible verdict artifacts as the contribution.

Submission type: **Late Breaking Abstracts**

Code: github.com/yangchunxuan/grown-grounded-language, tag v1.0-alife2026

The attack-machine

A candidate result is banked only if it passes *every* row of Table 1 simultaneously, with the verdict a committed, reproducible JSON. The *architecture-matched null* is the same architecture and parameter count with the mechanism ablated (an untrained random reservoir, a frozen-random channel); the *red-team* is a panel of independent agents each constructing a confound that explains the win without the

mechanism, and a surviving confound forces an added control or a retraction. The rig caught *six* of our own apparent wins as artifacts—e.g. a “+0.10” world-model denoising gain that an untrained reservoir of the same class beat by 3–4 $\times$, and a channel “intelligibility rise” that did not separate from a communication-blind control. The negatives below are therefore earned.

Table 1: The attack-machine: a result is banked only if every row passes.

Step	Required artifact	Pass criterion
Null	matched-architecture ablation	win-margin CI > 0 vs. null
Baseline	fair non-mechanistic baseline	treatment beats baseline
Seeds	≥ 6 seeds	bootstrap CI on the margin
Red-team	written confound report	no surviving shortcut
Verdict	committed JSON / log	reproducible from seeds
Replication	separate machine / model	same headline verdict

Five walls and a convergent diagnosis

Across five attacks, multiple independent agents, and repeated red-teams, no clean break survived to the headline (Table 2). Two convergent readings: in our tested regimes world design was a recurring bottleneck; and structure does not grow in worlds too small to require it—*installed* structure works. Each shortcut was first taken seriously and ruled out only when removing it from the world removed the wall. Per-wall apparent-win $\rightarrow$ confound $\rightarrow$ control $\rightarrow$ verdict chains, with the committed JSON artifacts, are documented in the repository.

Constructively closing the shortcuts

We build a world (a Relay-Tide Commons, RTC) that forecloses each shortcut by construction, and test the residue with the same rig (Figure 1).

Natural foraging. Grown-channel *content* is *not* load-bearing: any movement-prompt suffices because foraging en route makes message content irrelevant—a content-free shortcut at the survival layer.

Table 2: Each wall fell to a cheap shortcut a thin world permitted. Full apparent-win $\rightarrow$ confound $\rightarrow$ control $\rightarrow$ verdict chains and the committed verdict JSONs are in the repository (`WALLS-VERDICT.md`).

Wall	Cheap shortcut (the apparent win)	Ruled out by
Self-supervised prediction	temporal integration by generic recurrence (dead world)	untrained reservoir beats trained world-model
Within-life optimization	exhaustive coverage in a small state space	coverage cliff under a fixed budget
Compositional generalization	holistic codes under symmetric pressure	held-out joint stays 0 across eight attacks
Cooperation beyond communication	basin lottery / mutual-decode equivalence	selection-gate CI crosses 0
Spatial communication	co-location / flocking	content- vs. flocking-control comparison

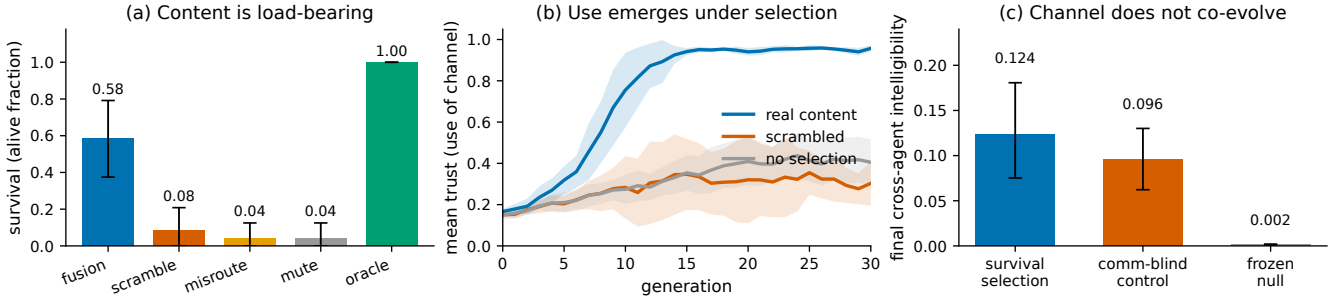


Figure 1: Results inside the constructed world; values from committed verdict artifacts, bars are bootstrap 95% CIs. **(a)** Lethal signaling (24 seeds): *fusion* (real content, 0.58) survives far above *scramble* (random, 0.08), *misroute* (correct content, broken content-to-location binding, 0.04) and *mute* (0.04); *oracle* (fenced) = 1.00. *fusion* beats each control by an exact paired McNemar test ($p < 5e-4$); the absolute-majority gate is inconclusive, so this is *relative*, and *misroute* shows the content-to-location *binding* (not signal presence) carries survival. **(b)** A heritable “trust” gene (8 seeds) rises to ≈ 0.96 under real content vs. ~ 0.3 – 0.4 under scrambled / no selection: *use* of a separately-trained channel is selected for (adoption, not de-novo emergence). **(c)** Co-evolving from random weights (6 seeds): the apparent intelligibility rise (0.124) does *not* separate from a communication-blind control (0.096; survival–control CI $[-0.05, 0.11]$ includes 0; underpowered), both far above the frozen null (0.002).

Lethal signaling (shortcuts foreclosed). With no local sensing and a forced lethal choice, grown-channel content—specifically its content-to-location binding—becomes load-bearing *relative* to degraded controls (Figure 1a); the decisive *misroute* control (correct content, broken binding) dies, so content, not signal presence, carries survival. Scope: relative not absolute, inside a narrow stakes window, in a world built to make content matter.

Use emerges under selection. A heritable trust gene gating message use rises from 0.17 to 0.96 under real content, versus ~ 0.3 – 0.4 under scrambled content or no selection (Figure 1b): selection for *using* a separately-trained channel (adoption), not de-novo emergence.

Does the channel co-evolve?—no detectable effect, underpowered. From random speaker/listener weights under survival selection alone (no truth gradient), even the strongest legitimate mechanism (tied encode/decode weights plus kin assimilation) shows no detectable growth of a load-bearing channel (Figure 1c): the apparent intelligibility rise did not separate from a communication-blind control ($n=6$, underpowered), consistent with clonal descent-

convergence rather than selection for communication.

Boundaries and contribution

The positives live inside constructed worlds; the core goal—a load-bearing grounded language that *grows* from scratch under selection—is not achieved. The recurring obstacle is world design plus a grounding/fidelity ceiling: a grown channel can carry information yet remain too imprecise, or too easily bypassed, to be load-bearing. We contribute (1) a transferable adversarial self-audit protocol that demonstrably catches false positives; (2) a precisely-mapped wall system; and (3) a constructive recipe (close shortcuts, then test) with scoped positives and a co-evolution null (no detectable effect, underpowered). We suggest emergent-communication claims be reported against architecture-matched nulls and communication-blind controls as standard. All code and verdict JSONs are committed and reproducible from fixed seeds; headline scripts were rerun from a fresh clone on a separate machine to the same verdicts, with an oracle-leakage audit. AI agents produced candidate implementations and adversarial critiques under this protocol.

References

- Chaabouni, R., et al. (2020). Compositionality and generalization in emergent languages. *ACL*.
- Foerster, J., et al. (2016). Learning to communicate with deep multi-agent reinforcement learning. *NeurIPS*.
- Geirhos, R., et al. (2020). Shortcut learning in deep neural networks. *Nature Machine Intelligence*.
- Kirby, S., Tamariz, M., Cornish, H., and Smith, K. (2015). Compression and communication in the cultural evolution of linguistic structure. *Cognition*.
- Lazaridou, A., and Baroni, M. (2020). Emergent multi-agent communication in the deep learning era. *arXiv:2006.02419*.
- Lazaridou, A., Peysakhovich, A., and Baroni, M. (2017). Multi-agent cooperation and the emergence of (natural) language. *ICLR*.
- Lehman, J., and Stanley, K. O. (2011). Abandoning objectives: evolution through the search for novelty alone. *Evolutionary Computation*.
- Lewis, D. (1969). *Convention*. Harvard University Press.
- Locatello, F., et al. (2019). Challenging common assumptions in the unsupervised learning of disentangled representations. *ICML*.
- Nowak, M. A. (2006). Five rules for the evolution of cooperation. *Science*.
- Ren, Y., et al. (2020). Compositional languages emerge in a neural iterated learning model. *ICLR*.
- Skyrms, B. (2010). *Signals: Evolution, Learning, and Information*. Oxford University Press.
- Steels, L. (2003). Evolving grounded communication for robots. *Trends in Cognitive Sciences*.